# STRAVYX

Australia's On-Demand Drone Services Network

## DJI Integration Catalogue - APIs, SDKs, Resources & Use Cases

Cloud API | Mobile SDK v5 | Payload SDK v3 | FlightHub 2 OpenAPI

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Source: docs/dji-integration-catalogue.md. Aligned to Master Business Summary (April 2026), executive-summary.md and mvp-build-timeline.md. A working catalogue of DJI developer APIs, SDKs, functions and features Stravyx can integrate, with descriptions and use-case scenarios.

Founder note: MVP (19 Jun) uses a manual Pilot-to-Cloud path (operator uploads raw data). DJI SDK integrations below are Phase 1+ (2026-2027) unless flagged otherwise. Reflects Cloud API v1.14.0 (Apr 2025) / PSDK 3.16.0 - verify endpoints against live DJI docs before build.

### 0. TL;DR - The Four DJI Pillars

| # | DJI product | Stravyx role | Phase |
|---|---|---|---|
| 1 | Cloud API (MQTT/HTTPS/WSS via Pilot 2 / Dock) | Backbone: telemetry, media return, dispatch, dock ops | Phase 1 |
| 2 | Mobile SDK v5 (MSDK) | Branded Stravyx operator flight app | Phase 1 (post-GA) |
| 3 | Payload SDK v3 (PSDK) | Custom payloads, onboard edge AI for Layer 2 | Phase 2 |
| 4 | FlightHub 2 OpenAPI / Modeling API / Terra | Layer 2 accelerator: photogrammetry, LiDAR, 3D | Phase 1-2 |

Key fact: with Cloud API, DJI drones/docks talk to our backend directly through DJI Pilot 2 (on the RC) or a DJI Dock - no mobile app required to get live telemetry, video, or automatic media upload. This is the single most important integration for the Stravyx tracker, dispatch, and Layer 1 -> Layer 2 handoff.

### 1. DJI Cloud API - The Integration Backbone

A cloud interface set that abstracts the drone/dock into an IoT 'thing model'. The gateway (DJI Pilot 2 on an RC, or a DJI Dock) connects to our cloud over MQTT (telemetry + commands), HTTPS (REST + media) and WebSocket (server push). Lets cloud-first platforms skip building a full MSDK app.

#### Two operating scenarios

- Pilot-to-Cloud - a human operator flies with DJI Pilot 2; we receive live data and push routes/media/video.
- Dock-to-Cloud - a DJI Dock runs autonomous missions; we schedule flights, monitor health, retrieve media, no human on site.

#### Supported hardware (Cloud API 1.14.0, Apr 2025)

- Pilot-to-Cloud: Matrice 4E/4T, M350 RTK, M300 RTK, M30 series, Mavic 3 Enterprise series.
- Dock-to-Cloud: DJI Dock 3 (M4D/4TD), DJI Dock 2 (M3D/3TD), DJI Dock (M30 series).

Access prerequisite: DJI developer account -> generate a Cloud API license -> bind Pilot 2 / Dock to our workspace. (One of the 'three DJI asks' in the commercial one-pager.)

#### MQTT topic model (how data flows)

| Topic pattern | Direction | Purpose |
|---|---|---|
| sys/product/{gw}/status | device -> cloud | Online/offline + update_topo (topology) |
| thing/product/{dev}/osd | device -> cloud | High-freq telemetry: GPS, battery, attitude, speed |

| | | |
|---|---|---|
| thing/product/{dev}/state | device -> cloud | Sparse state: firmware, payload, live-capacity |
| thing/product/{gw}/services | cloud -> device | Commands we invoke (start mission, photo, RTH) |
| thing/product/{gw}/events | device -> cloud | Device events: upload progress, HMS alarms |
| thing/product/{gw}/requests | device -> cloud | Device asks us for data (STS creds, org binding) |
| thing/product/{gw}/drc/{up,down} | both | Direct Remote Control (low-latency) |

REST module prefixes (/{module}/api/v1): manage (login, workspace, binding), wayline (routes + dispatch), media (fast-upload negotiation + callbacks), storage (temporary object-store STS), map, control.

### Cloud API functional modules - catalogue

| Module | Capability | Stravyx scenario |
|---|---|---|
| Device mgmt / topology | Discover + monitor docks/aircraft/payloads/RCs | Live fleet map (A-02); verified capability -> BEST MATCH SCORE |
| OSD telemetry | GPS, battery, attitude, speed stream | Live moving-pin consumer tracker (C-05); geofence audit |
| Live streaming | RTMP/RTSP/WebRTC video, start/stop | 'Watch live' for Immediate/Urgent security jobs; remote supervision |
| Wayline / route library | Upload/execute/report KMZ waylines | Scheduled tier + patrol blocks; weekly construction sweeps |
| Media library | STS fast-upload straight to S3 + callbacks | Auto Layer 1 -> Layer 2 handoff; kills manual upload (O-09) |
| HMS health mgmt | Real-time alarms/diagnostics | Dispatch safety gate; dock uptime SLA; CASA story |
| TSA situational awareness | Positions of nearby aircraft | Multi-operator metro conflict avoidance |
| Map elements | Sync pins/polygons/no-fly to Pilot 2 | Push customer site polygon to operator on win (M-05) |
| Dock control | Open/close, tasks, firmware, forced landing | Autonomous dock missions (Phase 1-2 vision) |
| DRC direct control | Low-latency real-time control | Stravyx-piloted dock takeover (Phase 2) |
| Org / device binding | Bind Pilot 2 / Dock to workspace | Operator onboarding as live data-backed link (O-01) |

## 2. DJI Mobile SDK v5 (MSDK) - The Operator App

Native Android/iOS SDK to build a fully custom flight app. Use when we want a branded Stravyx operator app with deeper control than Pilot 2 + Cloud API. Requires an MSDK app key. v5.3+ opened to consumer drones (Mini 3 / 3 Pro / 4 Pro) - lowering the equipment barrier for operator supply.

| Manager / API | Capability | Stravyx scenario |
|---|---|---|
| IWaypointMissionManager | Upload/execute/pause/resume KMZ missions | 'Fly this Stravyx mission' one-tap; auto status |
| IWPMZManager | Author/edit KMZ waylines on device | Operator fine-tunes supplied route for site |
| IVirtualStickManager | Programmatic control (+ obstacle avoidance) | Guided orbit/facade capture patterns |
| IMediaManager / DataCenter | List/preview/download/playback media | In-app raw preview + selective S3 upload |
| ILiveStreamManager | RTMP/RTSP live stream from app | Stream to customer/admin without Pilot 2 |
| Camera / gimbal control | Zoom, mode, shutter, gimbal angle | Enforce capture specs per category |
| Flight controller / telemetry | Position, battery, RTH, flight time | In-app tracker + safety gates |
| Simulator | Flight simulation | Operator training + QA of templates |
| KeyManager (product/firmware) | Read model/firmware/serial | Verified capability + compliance checks |

Supported drones (v5.x): M400, M4/M4D Enterprise, M350 RTK, M300 RTK, M30 series, Mavic 3 Enterprise, Mavic 3TA, plus Mini 3 / 3 Pro / 4 Pro. Roadmap: exec summary lists React Native (Expo) for native apps - plan a native module / bridge to RN, or a dedicated native operator app.

## 3. DJI Payload SDK v3 (PSDK) - Custom Payloads & Edge Compute

Firmware-level SDK (latest 3.16.0, Mar 2026) to build custom payloads via SkyPort V2 / X-Port / E-Port, talking to the aircraft's flight controller, GPS and transmission. Runs on Linux/RTOS/embedded and DJI Manifold onboard computers. Stravyx relevance: Phase 2 / strategic, not near-term.

- Onboard 'Layer 2 at the edge': run Stravyx AI (defect detection, counting) on Manifold 3 in flight -> partial insights before landing, lower cloud cost.
- Specialised deliverables: integrate gas / multispectral / LiDAR sensor for premium categories (agriculture, infrastructure).
- Megaphone / searchlight: active-response add-on for the Sentinel patrol product.

## 4. FlightHub 2 OpenAPI + Modeling API / DJI Terra - Layer 2 Accelerator

FlightHub 2 is DJI's cloud fleet-ops platform; its OpenAPI now exposes Modeling APIs so third parties can trigger cloud reconstruction: photogrammetry, LiDAR, and 3D Gaussian Splatting (shareable by QR, no login). DJI Terra (desktop) does 2D/3D/LiDAR-RGB/multispectral - but its legacy API is being phased out in favour of FlightHub 2 Modeling APIs. Build new integrations against FlightHub 2 OpenAPI.

| Capability | Stravyx Layer 2 scenario | Categories |
|---|---|---|
| Photogrammetry (2D ortho) | Auto site maps -> processed deliverable ($-06) | Construction, survey |
| 3D / 3DGS reconstruction | Digital twin / 3D deliverable, QR-shareable | 3D/twin, property |
| LiDAR reconstruction | High-accuracy point clouds, premium tier | Infrastructure, survey |
| Multispectral | Crop-health maps | Agriculture |

Scenario: replace the manual/placeholder Layer 2 accepted at GA ($-07) with real automated map/3D deliverables shortly after launch - unlocking the full ~40% blended margin sooner.

## 5. Where DJI Changes the Stravyx MVP Loop

| Step | MVP (manual, 19 Jun) | With Cloud API (Phase 1) |
|---|---|---|
| Dispatch | Broadcast brief; operator's own tools | Push KMZ wayline to Pilot 2 / dock |
| READY -> AIRBORNE | Operator taps + checklist | Auto from state/events; HMS safety gate |
| Consumer tracker (C-05) | Status timeline | Live position + live video |
| COMPLETE -> IN_REVIEW | Manual raw upload (O-09) | Auto media fast-upload; job auto-created |
| Layer 2 ($-05/06) | Manual/placeholder SLA | FlightHub 2 Modeling API reconstruction |
| Capability score | Operator self-declared tags | Verified from device topology/state |

Visibility rule reminder: DJI data flows into our backend, but the visibility module still enforces field-level filtering - operators/pilots/dock owners never see customer total or Layer 2 fee.

## 6. Phasing & Prioritisation (Recommended)

### Highest-leverage first (post-MVP)

1. Cloud API media auto-return - kills manual-upload friction in the core loop; small backend surface.
2. Cloud API OSD telemetry - live consumer tracker; big credibility/UX win, modest effort.
3. FlightHub 2 Modeling API - converts placeholder Layer 2 into real margin.
4. Cloud API wayline dispatch - unlocks Scheduled tier + patrol blocks (recurring revenue).
5. MSDK v5 app - branded operator experience once native apps are scoped.
6. Dock-to-Cloud + PSDK - the autonomous Phase 2 vision.

## 7. Access, Licensing & the 'Three DJI Asks'

| Requirement | For | How obtained |
|---|---|---|
| DJI developer account | All SDKs | developer.dji.com signup |
| Cloud API license | Cloud API (Pilot 2 / Dock) | Generated per workspace |
| MSDK app key | Mobile SDK app | Registered app on dev site |
| PSDK credentials + hardware | Payload dev | Dev kit + SkyPort/X-Port/Manifold |
| FlightHub 2 OpenAPI access | Modeling / fleet | developer.dji.com/flighthub-api |
| Enterprise relationship | Volume, support, co-marketing | Credibility package -> DJI Enterprise intro |

Ecosystem scale (for one-pager): 100,000+ DJI developers; 750+ cloud platforms on Cloud API since Mar 2022; 110+ PSDK payloads mass-produced. Positions Stravyx as a credible cloud-platform partner.

### Suggested three DJI asks (one-pager, K-02)

1. Cloud API + FlightHub 2 OpenAPI enablement for the Stravyx workspace (Pilot-to-Cloud first).
2. Enterprise hardware/partner pathway for docks (Dock 2/3), tied to Stravyx Finance (4K Group).
3. Co-marketing / reference status as an Australian drone-services cloud platform.

## 8. Open Questions to Validate Before Build

- Confirm current Cloud API version + exact module endpoints (this reflects v1.14.0, Apr 2025).
- Confirm FlightHub 2 Modeling API commercial terms, quotas, output formats vs own pipeline (Roboflow/Pix4D).
- Decide MSDK-native vs RN bridge for the operator app (affects Phase 1 mobile scoping).
- Networking: DJI gateways need outbound MQTT/HTTPS/WSS to our endpoints - confirm no dock-site firewall blockers.
- Data residency: Cloud API media into S3 ap-southeast-2; confirm DJI region routing meets AU expectations.

Markdown source: docs/dji-integration-catalogue.md. Related: executive-summary.md, mvp-build-timeline.md. Source: DJI Developer (developer.dji.com), Cloud-API-Doc, MSDK v5 & PSDK references, FlightHub 2 OpenAPI - verify before implementation.